Saussure in HLLSet Algebra

Part III: Diachronic Linguistics

A Computational Exegesis of the *Course in General Linguistics*, Part Three, in the Boolean Lattice and GF(2) Ring of HLLSet Algebra

Third paper in the series *Saussure in HLLSet Algebra*

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Abstract

Part III of the series releases the exclusion that Part II imposed: time. Part Three of the *Course*, *Diachronic Linguistics*, is re-founded as the theory of the **pivot**. The center of the exegesis is the generation change of the basis frame. A phonetic change is a pivot of the ring—an *extension* (a new opposition is born) and a *rotation* (old oppositions are re-pivoted)—and its "absolute regularity" is the determinism of the Gaussian step, while its "blindness" is the fact that the pivot reads only bit positions, never values. Analogy, the counterpart of sound change, is the *fourth proportional of GF(2)*: given three nodes A, B, C of the span, the analogical fourth is $D = A\Delta B\Delta C$, the unique node with $A\Delta B = C\Delta D$; it creates inside the span and never moves the basis. Folk etymology is a *false* fourth proportional, whose error the algebra measures exactly as $C\Delta C'$. Agglutination is the promotion of a *view* to an *original*: a frequently welded join is re-ingested as a new unit, and—because the span is closed under Δ but not under $\cup$— agglutination is the second source of basis growth. Diachronic units, identities, and realities close the part: a diachronic unit is a path of values across generations, diachronic identity is IICA persistence, and diachronic reality is the recorded transition. The paper closes with the complete dictionary of Part Three and the residual handed to Part IV: geographical diversity.

Keywords: Saussure; diachronic linguistics; sound change; phonetic change; analogy; folk etymology; agglutination; diachronic unit; diachronic identity; pivot; basis frame; generation; change of basis; GF(2) ring; Boolean lattice; HLLSet; computational exegesis.

1 Introduction

1.1 Where Part III stands in the series

Part I of this series [5] re-founded Part One of the *Course*—the sign, arbitrariness, linearity, immutability and mutability, and the static/evolutionary distinction—as theorems and invariants of HLLSet Algebra. Part II [6] re-founded Part Two, Synchronic Linguistics, as the differential theory of value: relative to a basis frame $F = (B_1, \ldots, B_k)$, the hard key $c \in \mathrm{GF}(2)^k$ is the exact compression of a node's value, identity and value coincide on the span, and "only differences, without positive terms" became the completeness and difference theorems of [6]. Both papers held one variable fixed: *the frame*. Part II said so explicitly: the spill was treated as a boundary, not as a cause, and the residual handed to Part III was named—the pivot itself [6].

Part Three of the *Course*, *Diachronic Linguistics*, is where Saussure lets time in. Its subject is not relations between coexisting terms but relations between *successive* terms: sound changes, their grammatical consequences, analogy, folk etymology, agglutination, and the diachronic counterparts of units, identities, and realities. Part III of this series re-founds each of

those chapters in the algebra, around a single executable event: the **pivot**, the recomputation of the basis frame that increments the generation γ and re-coordinates every cached value. The thesis of this paper is one sentence: **diachronic linguistics is the theory of the pivot and its two species—the blind pivot (sound change) and the exact non-pivot (analogy)**. Everything else in Part Three is a corollary of that distinction.

1.2 Method, unchanged

The method is that of Parts I and II [5, 6]: no metaphor, principles as theorems or invariants, and ambiguity recorded as data. The translation note of Part I applies unchanged: we cite Roy Harris's translation [2], keep the standard pair *signifier/signified* (Harris: *signal/signification*), and retain the Russian edition [4] for verification. References to the *Course* are by part, chapter, and section.

1.3 What Part Three contains, and the roadmap

Part Three of the *Course* has eight chapters: (I) general observations on the diachronic perspective; (II) sound changes; (III) grammatical consequences of phonetic evolution; (IV) analogy; (V) analogy and evolution; (VI) popular etymology; (VII) agglutination; (VIII) diachronic units, identities, and realities, with the appended discussion of subjective and objective analysis folded into Chapter VIII here.

Section 2 fixes the standing definitions inherited from Parts I and II and adds the diachronic layer: transitions, pivots, and the change-of-basis transport. Section 3 re-founds the general observations (the diachronic perspective as the trajectory). Section 4 is the center: sound change as the pivot, regularity as determinism, blindness as bit-level reading. Section 5 derives the grammatical consequences (bonds broken by transport). Section 6 is the second center: analogy as the fourth proportional of GF(2). Section 7 separates the two forces of evolution. Section 8 re-founds folk etymology as a false fourth proportional. Section 9 re-founds agglutination as the promotion of a view to an original. Section 10 closes with diachronic units, identities, and realities. Sections 11–13 collect the dictionary, state what Part III adds, and hand the geographical residual to Part IV.

2 Preliminaries: the synchronic inheritance and the diachronic layer

We restate, with their labels, the standing definitions of Parts I and II; proofs and motivation are in [5, 6]. We then add the diachronic layer: transitions, pivots, and transport.

Definition 2.1 (Plane, lattice, ring)**.** *The **plane** is $\mathbb{B} = \{0, \dots, 32767\}$; the **HLLSet lattice** is $\mathcal{B} = \mathcal{P}(\mathbb{B})$, a Boolean lattice under $\cup, \cap, \subseteq$ and a Boolean ring under Δ (written Δ) and $\cap$. Cardinalities are $|X|$; the complement of X in $\mathbb{B}$ is $\overline{X}$.*

Definition 2.2 (IICA morphism; ingestion; materialization)**.** *$\mu : \mathcal{T} \to \mathcal{B}$ is an **IICA morphism** if it is idempotent (deterministic), immutable, and content-addressed; it extends to finite token sets by $\mu(X) = \bigcup_{t \in X} \mu(\{t\})$. **Materialization** $\rho : \mathcal{B} \dashrightarrow \mathcal{C}$ is the partial inverse that restores an ordered token context through the door [10].*

Definition 2.3 (BSS; frames; generations; keys; spill)**.** *Against a reference $B_w \subseteq \mathbb{B}$, the **directed soft key** is $\mathsf{BSS}_w(X) = |X \cap B_w|/|B_w|$. A **basis frame** is a row-echelon GF(2) basis $F = (B_1, \dots, B_k)$ with distinct leading bits, spanning a set of observed nodes, and stamped with a **generation** γ that increments on recomputation. Relative to F, each $X \in \langle F \rangle$ has a unique **hard key** $c \in \mathrm{GF}(2)^k$ with*

$$X = \bigoplus_{i : c_i = 1} B_i,$$

*the **soft key** $w_i = \mathsf{BSS}_{B_i}(X)$ for $i = 1, \ldots, k$, and the **spill** $s = |X \setminus \mathrm{cover}(F)|$, where $\mathrm{cover}(F) = \bigcup_i B_i$ [9].*

Definition 2.4 (Synchronic fiber; diachronic trajectory)**.** *The **synchronic fiber** at time t is $\mathsf{Syn}(t) = (\mathcal{B}, F(t), \gamma(t))$; the **diachronic structure** is the family $\mathsf{Dia} = \{\mathsf{Syn}(t) : t \in T\}$ with the recorded transitions between consecutive fibers.*

Definition 2.5 (The sign)**.** *The **sign** is the pair $\mathcal{S} = (\mu, \rho)$; for a node $A \in \mathcal{B}$, the signifier is the bit-set A and the signified is its differential value plus the materialized context $\rho(A)$ [5, 6].*

2.1 The diachronic layer

Part III needs three objects that Parts I and II used but did not isolate: the transition, the pivot, and the transport.

Definition 2.6 (Transition; the D/R/N triad)**.** *Let $H(t{-}1)$ be the committed base of the previous turn and $S(t)$ the state after the join of turn t. The **recorded transition** is*

$$\mathsf{Tr}(t) \;=\; \big(\, S(t),\; H(t{-}1),\; \mathsf{Dep},\; \mathsf{Ret},\; \mathsf{New} \,\big),$$

where the three components are the sets

$$\mathsf{Dep} \;=\; H(t{-}1) \setminus S(t), \qquad \mathsf{Ret} \;=\; H(t{-}1) \cap S(t), \qquad \mathsf{New} \;=\; S(t) \setminus H(t{-}1),$$

*the **departed**, **retained**, and **new** parts of the transition; they partition $H(t{-}1) \cup S(t)$. The **coverage** of the transition is $\mathrm{cov}(t) = |\mathsf{Ret}|/|S(t)|$ when $|S(t)| > 0$.*

Remark 2.7 (One letter, two roles, now separated)**.** *Following the EWM series, the letters $D/R/N$ are reserved for the transition decomposition of Definition 2.6 (departed/retained/new) [9]. The reading of a context X against a document D—the numbers $N_D = |D \setminus X|$, $R_D = |D \cap X|$, $\mathrm{cov} = R_D/|D|$—is called the* coverage reading *throughout the series, in Parts I and II as here. The two roles share one letter no longer.*

Definition 2.8 (Pivot: extension and rotation)**.** *Let $F(t{-}1)$ be a frame of generation $\gamma(t{-}1)$, and let Y be the original HLLSet of turn t. Reduce Y against $F(t{-}1)$ by one Gaussian step, obtaining the **residual** $\mathrm{res} = \mathrm{res}_{F(t-1)}(Y)$.*

- *If $\mathrm{res} = \varnothing$, the turn is an **in-span event**: $F(t) = F(t{-}1)$ and $\gamma(t) = \gamma(t{-}1)$.*
- *If $\mathrm{res} \neq \varnothing$, the turn is a **pivot**. Let p be the leading bit of* res*. **Extension** adds $B_{k+1} = \mathrm{res}$ as a new basis vector with pivot p; **rotation** replaces every existing basis vector B_i that contains p by $B_i \Delta \mathrm{res}$. The new frame $F(t)$ is the resulting row-echelon basis with distinct leading bits, and $\gamma(t) = \gamma(t{-}1) + 1$.*

Proposition 2.9 (The pivot is exact)**.** *Let $F(t{-}1)$, Y, and* res *be as in Definition 2.8.*

1. *$\mathrm{res} \neq \varnothing$ if and only if $Y \notin \langle F(t{-}1) \rangle$.*
2. *After a pivot, $\dim\langle F(t) \rangle = \dim\langle F(t{-}1) \rangle + 1$; the old span is contained in the new span, and every old basis vector keeps unique coordinates in $F(t)$.*
3. *The pivot map $(F, Y) \mapsto F'$ is deterministic: in a fixed coordinate order of the plane, the same basis and the same incoming bits always produce the same extension and the same rotation.*
4. *Spill $|Y \setminus \mathrm{cover}(F(t{-}1))| > 0$ implies $\mathrm{res} \neq \varnothing$, but the converse can fail: a node may lie inside the cover yet outside the span (the meet obstruction [9]). The exact trigger of the pivot is $\mathrm{res} \neq \varnothing$, not positive spill.*

Proof. (1) is the definition of the residual: the remainder vanishes exactly for span elements. (2) The leading bit p of res is, by construction, not a leading bit of $F(t{-}1)$, so res is linearly independent of the old basis; the span grows by exactly one, and monotonicity of the span gives

the containment. (3) The Gaussian step and the rotation rule are deterministic functions of the bit patterns involved. (4) Every span element is a subset of cover(F) (an XOR of subsets is a subset of their union), so a bit outside the cover forces $Y \notin \langle F \rangle$; the converse fails because $B_i \cap B_j \subseteq \text{cover}(F)$ while $B_i \cap B_j \notin \langle F \rangle$ in general. □

Definition 2.10 (Change-of-basis transport)**.** *Let $F(t{-}1) = (B_1, \ldots, B_k)$ and $F(t) = (B'_1, \ldots, B'_{k'})$ be two frames of a monotone ring, so $\langle F(t{-}1) \rangle \subseteq \langle F(t) \rangle$. The* ***change-of-basis matrix*** $\mathbf{M} = \mathbf{M}(t)$ *is the $k \times k'$ matrix over* GF(2) *whose i-th row is the coordinate vector of B_i in $F(t)$. A hard key c at $t{-}1$* ***transports*** *to the hard key*

$$c' \;=\; c\mathbf{M}$$

at t (row-vector convention), and this transport is exact.

Proposition 2.11 (Exact transport)**.** *With the notation of Definition 2.10, if $X = \bigoplus_{i:c_i=1} B_i$, then $X = \bigoplus_{j:c'_j=1} B'_j$ with $c' = c\mathbf{M}$. Consequently the coordinates of a fixed node change with the frame, but the change is a deterministic linear map recorded in the ring, and every reading is valid relative to the generation in which it was taken.*

Proof. Each old basis vector expands uniquely in the new basis; substituting these expansions into X and collecting coefficients gives $c' = c\mathbf{M}$ by linearity over GF(2). □

3 Chapter I: General observations

Saussure opens Part Three by delimiting the diachronic perspective: it studies relations between *successive* terms that replace one another in time, not relations between coexisting terms; its facts are events, not states [2, Part III, ch. I].

Reading 3.1 (The diachronic perspective). The relations between coexisting terms are the fiber readings of Part II (hard keys, soft keys, containment). The relations between successive terms are the recorded transitions $\mathsf{Tr}(t)$. The diachronic perspective is the reading of the trajectory Dia through its transitions, exactly as the synchronic perspective is the reading of one fiber $\mathsf{Syn}(t)$.

Observation 3.2 (Events are not states). A synchronic fact is a state: it can be read, and re-read, inside one fiber. A diachronic fact is an event: it happens once, between two fibers, and is recorded. The two perspectives of Part I—the axis of simultaneities and the axis of successions—are not two ways of looking at the same object; they are two different types of object. The algebra makes this a type distinction: $\mathsf{Syn}(t)$ is a triple $(\mathcal{B}, F(t), \gamma(t))$; $\mathsf{Tr}(t)$ is a five-tuple of nodes with a direction.

Proposition 3.3 (The diachronic law is the recorded transition)**.** *Each transition $\mathsf{Tr}(t)$ is* imperative*—it happened, and the commit DAG records it—but* not general*—no identity of the lattice forces it, and it holds of that transition alone. This is Part I's Theorem on the two kinds of law [5], restated with the transition record made explicit: the diachronic laws are exactly the recorded transitions, and the synchronic laws are exactly the identities of the fiber.*

4 Chapter II: Sound changes as pivots

Saussure's chapter on sound changes makes two claims that have puzzled readers. First, sound changes are *absolutely regular*: a sound change is not an isolated accident but a law holding of every occurrence of the sound in question, under stated conditions. Second, sound changes are *blind*: they alter the material of words without any regard for grammar or meaning [2, Part III, ch. II].

Reading 4.1 (Sound change is a pivot). A sound change is a pivot of Definition 2.8: a turn whose original Y is not in the span forces the frame to recompute. The *extension* is the birth of a new opposition (a new distinctive coordinate); the *rotation* is the re-pivoting of the old oppositions that contain the new pivot bit. The generation γ increments, and every hard key in the span is re-coordinated.

Observation 4.2 (A sound change is local in the plane, global in the frame). Let X be a signifier before the change and X' the same signifier after. A sound change replaces one sound by another; in the algebra the replaced sound was mapped by μ to one bit b, and the new sound is mapped to a different bit $b' \notin X$. The surrounding bits—everything the previous state of the HLLSet had set around b—are retained, so

$$X' = (X \setminus \{b\}) \cup \{b'\}, \qquad X \Delta X' = \{b, b'\}, \qquad \mathsf{BSS}_X(X') = \frac{|X| - 1}{|X|} = 1 - \frac{1}{|X|}.$$

The two versions are therefore almost identical as lattice nodes: their symmetric difference has cardinality 2, and their directed similarity is $1 - 1/|X|$, arbitrarily close to 1 as the node grows. The minor bit change survives at the lattice level: the new form stays in the same neighbourhood of the Boolean lattice, and the retained bits keep it co-occurring with its old context. Yet these same two bits, when b' lies outside the span, force a pivot: the generation increments and every hard key in the fiber is re-coordinated. A sound change is thus local in the plane—two bits—and global in the frame—all values. It is the chess move of Part I made literal: one piece moves, the whole position is re-valued.

Proposition 4.3 (Phonetic change as extension and rotation)**.** *Let* $\mathrm{res} \neq \varnothing$ *be the residual of turn* t*. Then the pivot acts as follows. (i)* Extension: *the span dimension grows by one; the residual becomes a new basis vector, a new distinctive opposition. (ii)* Rotation: *exactly the old basis vectors containing the new pivot bit are replaced by their XOR with the residual; the others are untouched. The stored nodes are never rewritten—IICA immutability—so a sound change is a change of the frame around the signs, never of the signs themselves.*

Proof. This is the Gaussian step of [9], restated in the language of Definition 2.8. The rotation count is the number of old oppositions that re-index themselves through the new one; it is zero exactly when no old basis vector contains the pivot bit. □

Proposition 4.4 (Regularity and blindness, re-founded)**.** *(i)* Absolute regularity. *The pivot map of Proposition 2.9 is a function: in a fixed coordinate order, the same basis and the same incoming bits always produce the same extension and rotation. Saussure's "absolute regularity" is therefore determinism of the Gaussian step. (ii)* Blindness. *The pivot reads only bit positions—the leading bits, the membership of the pivot bit in old vectors, the XOR rule. It never consults the LUT, the materialized context, the hard keys, or any value. A sound change is blind because it is a computation on the plane, not on the system of values.*

Remark 4.5 (The trigger of sound change is a residual)**.** *The exact trigger of a sound change is* $\mathrm{res} \neq \varnothing$ *(Proposition 2.9(4)), stated uniformly in Part I [5] and here. A node with positive spill is visibly new (bits no opposition can express); a node inside the cover but outside the span—the meet obstruction—is invisibly new, and it pivots just the same. Saussure's sound changes have the same two faces: some are visible to speakers (a new sound appears), others are visible only to the system (a distinction reorganizes without new material).*

Remark 4.6 (The cost of a sound change)**.** *The rotation count and rotation mass of [9] are the exact measures of a sound change's cost: how many oppositions had to re-index themselves through the new one. A change with rotation zero adds a new distinction without disturbing any old one—the diachronic analogue of a purely additive sound. A change with large rotation is the diachronic analogue of a systematic shift: many old distinctions are re-drawn at once.*

Object	Before the pivot ($F(0)$, $\gamma = 0$)	After the pivot ($F(1)$, $\gamma = 1$)
Plane	$\{0,1,2,3\}$	
Basis	$B_1 = \{0,1\},\ B_2 = \{1,2,3\}$	$B_1 = \{0,1\},\ B_3 = \{0,2\},\ B_2' = \{0,1,3\}$
Incoming original	$Y = \{0,2\}$	residual res $= Y = \{0,2\}$, pivot bit 2
Extension	—	$B_3 = \text{res}$
Rotation	—	$B_2' = B_2 \Delta \text{res} = \{0,1,3\}$
Node $X = B_1 \Delta B_2 = \{0,2,3\}$	hard key $(1,1)$	hard key $(1,1,1)$
Transport	—	$B_1 \mapsto (1,0,0),\ \ B_2 \mapsto (0,1,1)$

Table 1: A miniature sound change on a four-bit plane. The pivot extends the basis by the residual and rotates the one old basis vector containing the new pivot bit; the stored node X is untouched while its hard key changes from $(1,1)$ to $(1,1,1)$.

5 Chapter III: Grammatical consequences of phonetic evolution

Saussure's chapter III traces what sound change does to grammar: it breaks grammatical bonds, produces alternations, and never creates new grammatical forms—only destroys or re-arranges [2, Part III, ch. III].

Reading 5.1 (Bonds are coordinate coincidences). In Part II, the associative series of a node X is the set of nodes sharing a coordinate with it, and syntagmatic solidarities are coverage guards. A *grammatical bond* between two nodes is a shared coordinate in the current frame. The pivot re-coordinates the whole span, so bonds break or form without any node being rewritten.

Proposition 5.2 (Bonds under transport)**.** *Let $X, Y \in \langle F(t-1) \rangle$ have hard keys c, d, and let $\mathbf{M}$ be the change-of-basis matrix of the pivot. In the new frame, X and Y share coordinate j exactly when*

$$(c\,\mathbf{M})_j \;=\; (d\,\mathbf{M})_j.$$

A pivot therefore breaks a bond (X, Y, i) whenever $c_i = d_i$ but $(c\mathbf{M})_j \neq (d\mathbf{M})_j$ for every j; and it creates a bond whenever $c_i \neq d_i$ for all i but some transported coordinates agree. The grammatical consequences of sound change are exactly these breakings and creations, computed by one matrix multiplication.

Proof. Immediate from Proposition 2.11: the new coordinates are $c\mathbf{M}$ and $d\mathbf{M}$, and coordinate agreement is equality of the corresponding entries. □

Corollary 5.3 (Alternation is one node, two generation-stamped values)**.** *An alternation—two forms of "the same" unit in complementary contexts—is a single stored node whose value changed under a pivot, read against two generation stamps. The node is one (IICA); its coordinates are two (transport). This is the sheet of paper of Part II [6] moving through time: one cannot change the value without changing the sign's position in the system, and one cannot change the stored node at all.*

Remark 5.4 (Sound change destroys, never builds)**.** *Saussure insists that phonetic evolution never creates grammatical forms. The algebra gives the reason: a pivot only re-coordinates existing nodes; it introduces no new node. Creation is the work of the other force, analogy, to which we now turn.*

6 Chapter IV: Analogy as the fourth proportional

Saussure's chapter IV introduces the second diachronic force. Analogy needs a model and its regular imitation: it takes *three* terms to create a *fourth*, by the proportion

$$\text{orātor} : \text{orātōrem} \;=\; \text{honōs} : \text{honōrem},$$

and the created form is a grammatical, intelligent construction—unlike sound change, which is blind [2, Part III, ch. IV].

Reading 6.1 (Analogy is the parallelogram law). The analogical proportion $A : B :: C : D$ is read as the equality of differences

$$A\Delta B \;=\; C\Delta D.$$

The proportion is then an equation in the Boolean ring, and it has a unique solution for the fourth term: $D = A\Delta B\Delta C$.

Definition 6.2 (Analogical fourth). *For nodes $A, B, C \in \mathcal{B}$, the **analogical fourth** is*

$$\mathsf{Ana}(A, B, C) \;=\; A\Delta B\Delta C.$$

*The proportion $A : B :: C : D$ **holds** exactly when $A\Delta B = C\Delta D$, equivalently $D = \mathsf{Ana}(A, B, C)$.*

Theorem 6.3 (The analogy theorem). *Let F be a frame and $A, B, C \in \langle F\rangle$ with hard keys a, b, c. Then:*

1. Existence and uniqueness. *$D = \mathsf{Ana}(A, B, C)$ is the unique node of $\mathcal{B}$ with $A\Delta B = C\Delta D$.*
2. Closure. *$D \in \langle F\rangle$, with hard key $d = a\Delta b\Delta c$ (coordinatewise XOR).*
3. Symmetry. *The proportion is its own mirror: also $A\Delta C = B\Delta D$ holds.*
4. Conservatism. *The frame is untouched: γ does not increment, and the span does not change. Analogy creates a node; it creates no opposition.*

Proof. (1) In the group $(\mathcal{B}, \Delta)$, the equation $A\Delta B = C\Delta D$ has the unique solution $D = A\Delta B\Delta C$ (add C to both sides; in GF(2), $C\Delta C = \varnothing$). (2) The span is a vector subspace over GF(2), so it is closed under XOR; expanding in the basis gives $d = a\Delta b\Delta c$. (3) $A\Delta C = A\Delta C\Delta(B\Delta B) = (A\Delta B)\Delta(B\Delta C)$, while $B\Delta D = B\Delta A\Delta B\Delta C = A\Delta C$. (4) No pivot is involved: D is computed from the existing basis, so F and γ are unchanged. □

Proposition 6.4 (Analogy and sound change are type-distinct). *Sound change is a* basis event*: the span changes, the generation increments, stored nodes are untouched, and their values change. Analogy is a* span event*: the span is untouched, the generation is unchanged, and a new node appears with a value in the existing system of oppositions. Saussure's "analogy is the counterpart of phonetic change" is, in the algebra, a type distinction between the only two kinds of diachronic event.*

Remark 6.5 (Why analogy feels intelligent). *Analogy is the one diachronic operation that uses the system of values itself—the hard keys—as its raw material. A pivot uses only bits; the analogical fourth uses coordinates. Intelligence, in Saussure's sense, is exactly this: the operation consults the relations, not the substance.*

7 Chapter V: Analogy and evolution

Saussure's chapter V asks how analogy, which is a grammatical and conscious procedure, can be a factor of *evolution*—a force of change like sound change [2, Part III, ch. V].

Proposition 7.1 (Analogy is conservative in the system, innovative in the lexicon)**.** *Let* $D = \mathsf{Ana}(A, B, C)$ *be an analogical creation. Then* $\mathrm{res}_F(D) = \varnothing$ *and* $|D \setminus \mathrm{cover}(F)| = 0$ *for the frame in which* A, B, C *lie: an analogical creation never forces a pivot, never adds a bit outside the frame cover, and never changes the system of oppositions. It is* conservative *in the system—it reuses the community's existing coordinates—and* innovative *in the lexicon—it names a node that may never have been uttered before. Once the creation is used in parole and committed, it enters the record as a new original; but the frame, and with it the grammar, stays exactly as it was.*

Proof. Theorem 6.3(2) gives $D \in \langle F \rangle$; hence the residual is empty. Every span element is a subset of the cover, so the spill is zero. Committing D adds a node to the store, not a direction to the span. □

Proposition 7.2 (The two forces decompose the record)**.** *Every diachronic event is of exactly one of two kinds: a* pivot *(Definition 2.8), which increments the generation and changes the system of oppositions; or an* in-span commit*, which records a node without changing the frame. Sound change is a pivot; analogy is an in-span commit; agglutination (Section 9) can be either, depending on whether the weld escapes the span. The commit DAG labels each commit with its generation increment, so the two forces are separable in the record itself.*

Remark 7.3 (The speaker's only creative act)**.** *Part I re-founded "the language changes although the speakers do not change it" as a two-layer observation [5]. Part III now locates the speaker's contribution precisely. Speakers do not pivot (the pivot is the system's response to residue); speakers* do *analogize—and even then they build with the frame the community already shares. The individual creative act is real, and it is exactly as conservative as Saussure said.*

8 Chapter VI: Popular etymology as a false fourth proportional

Saussure's chapter VI treats folk etymology: an opaque form is re-analyzed by being compared, wrongly, with something familiar. Unlike analogy, it is isolated and limited in scope, and it rests on an incorrect evaluation of the old form [2, Part III, ch. VI].

Reading 8.1 (Folk etymology is a false analogy)*.* A folk etymology is an analogy computed from a *mis-identified* term: the speaker takes the triple (A, B, C') where C' is not the true third term C but a node confused with it by soft-key resemblance. The result is the false fourth proportional $D' = \mathsf{Ana}(A, B, C')$.

Definition 8.2 (False fourth proportional)**.** *Let* C *be the true third term and* C' *a node mis-identified with* C*. The* ***folk-etymological creation*** *is*

$$D' \;=\; \mathsf{Ana}(A, B, C') \;=\; A \Delta B \Delta C'.$$

Its ***error*** *is the symmetric difference between the false and the true fourth:*

$$D \Delta D' \;=\; (A \Delta B \Delta C) \Delta (A \Delta B \Delta C') \;=\; C \Delta C'.$$

Proposition 8.3 (The algebra measures the incorrect evaluation)**.** *(i)* $D' = D$ *if and only if* $C' = C$*: the folk etymology is exact exactly when the identification is exact, and then it simply is analogy. (ii) The error of the creation equals the error of the identification:* $D \Delta D' = C \Delta C'$*. (iii) The false creation carries no provenance support for the re-analysis: the recipe's sources are the nodes it actually uses, and the historical relation it claims is not among them.*

Proof. (i) and (ii) are immediate from the group law of Δ. (iii) A view records the provenance of the expression actually computed; the claimed relation to C appears in no source, so the objective analysis (Section 10) does not confirm it. □

Remark 8.4 (Why folk etymology is isolated)**.** *Analogy is systematic because it is computed in the span, with the shared coordinates: the same proportion holds for every node with the same keys. Folk etymology is isolated because it is computed from a soft-key confusion that is local to one form: the mis-identification $C' \approx_{\mathsf{BSS}} C$ need not transfer to any other triple. Saussure's limitation of scope is, in the algebra, the non-functoriality of the confusion map.*

9 Chapter VII: Agglutination as the promotion of a view

Saussure's chapter VII treats agglutination: two elements that occur side by side in the chain merge into one unanalyzable unit. It works only within syntagmas, it is involuntary, and together with analogy it is the main source of new forms [2, Part III, ch. VII].

Reading 9.1 (Agglutination welds a view into an original). In the EWM storage discipline [9], only originals and base HLLSets are materialized; every compound is a *view*—a recipe plus provenance, recreated on read. Agglutination is the event that breaks the discipline on purpose: a compound that the chain keeps producing is promoted to an *original*—the join is registered as a stored node with its own content address, and thereafter it behaves as a unit.

Definition 9.2 (Weld)**.** *Let $A, B \in \mathcal{B}$ be nodes whose co-occurrence in materialized contexts is systematically high (the retention reading of the D/R/N triad, accumulated over windows). The* ***weld*** *of A and B is the registration of*

$$O \;=\; A \cup B$$

as a new original: a node that was previously only a view expression `union` *over sources A, B now has its own stored bits and its own content address.*

Proposition 9.3 (Agglutination can pivot)**.** *Let $A, B \in \langle F \rangle$. Then*

$$A \cup B \;=\; A \Delta B \Delta (A \cap B),$$

so the weld $O = A \cup B$ lies in the span if and only if the meet $A \cap B$ lies in the span. A weld of in-span nodes therefore forces a pivot exactly when the meet is not representable. *Agglutination is thus the second source of basis growth: sound change pivots by residue, agglutination pivots by meet.*

Proof. The identity $A \cup B = A \Delta B \Delta (A \cap B)$ holds in any Boolean ring. Since $A \Delta B \in \langle F \rangle$ (span closed under XOR), membership of the union in the span is equivalent to membership of the meet; the rest is Proposition 2.9(1). □

Remark 9.4 (Three forces, two of which move the basis)**.** *The diachronic forces now have exact types. Sound change: a pivot by residue, blind, moves the basis. Agglutination: a pivot by meet, syntagmatic, involuntary, moves the basis when the weld escapes the span, and records an in-span original otherwise. Analogy: never a pivot, associative, intelligent, populates the lexicon within the span. Saussure's chapter VII observation that agglutination, unlike analogy, works only within syntagmas is the type statement: agglutination is a join (a chain operation), analogy is a XOR (a frame operation).*

10 Chapter VIII: Diachronic units, identities, realities

Saussure closes Part Three with the diachronic counterparts of Part Two's units, identities, and realities, and the appended distinction between subjective and objective analysis [2, Part III, ch. VIII].

Reading 10.1 (Diachronic unit, identity, reality). The synchronic unit is a node read in one fiber. The diachronic unit is the *path* of that node's values across generations. Diachronic identity is the persistence of the node itself; diachronic reality is the recorded transition.

Definition 10.2 (Diachronic unit)**.** *A* ***diachronic unit*** *is a pair* $(X, [t_1, t_2])$*: a stored node* X *together with the interval of generations over which it is referenced, read as the value trajectory*

$$c(t) \;=\; \textit{hard key of } X \textit{ in } F(t), \qquad t_1 \le t \le t_2,$$

with the transport matrices $c(t{+}1) = c(t)\mathbf{M}(t{+}1)$ *connecting consecutive values. It is a path in the trajectory* Dia*, not a node of the lattice.*

Proposition 10.3 (Diachronic identity is IICA persistence)**.** *A node at* t_1 *and a node at* t_2 *are diachronically identical if and only if they are the same bit-set,*

$$X(t_1) \;=\; X(t_2).$$

Their values may differ: the hard keys $c(t_1)$ *and* $c(t_2)$ *are equal if and only if no pivot between* t_1 *and* t_2 *moved them, i.e.* $c(t_1)\mathbf{M} = c(t_1)$*. Identity is content; value is coordinates. The distinction of Part II—identity and value coincide within one fiber—now has its diachronic counterpart: across fibers, identity persists while value shifts.*

Proof. The first assertion is content addressability (IICA): the same bits name the same node, and different bits name different nodes. The second is Proposition 2.11: values change exactly by the transport matrices of the intervening pivots. □

Proposition 10.4 (Subjective and objective analysis)**.** *The* ***subjective analysis*** *of a sign is its reading in the current frame: the soft key and the hard key of* $F(t)$*—what a speaker of the current system can see. The* ***objective analysis*** *is the provenance chain: the view recipe, the commit DAG, and the transport matrices—what the record shows. A pivot separates the two: subjectively, "the same word" has simply changed its value; objectively, the change is a recorded transition with a generation stamp. The algebra keeps both, and falsifies neither.*

Remark 10.5 (The residual of Part Three)**.** *Part Three leaves one residual, and it is the same one Saussure leaves: the* cause *of sound change. The algebra records that a pivot happened, how many oppositions it rotated, and what it did to every value; it does not, and cannot, say why the residue arrived—why this community, at this time, produced this new material. The trigger is parole (the external stream), and parole is outside the system by construction. The algebra turns Saussure's "sound changes have no internal cause" into a type boundary: the pivot is inside the system; its occasion is not.*

11 The dictionary of Part Three

12 What Part III adds to Parts I and II

Part I named the generation mechanism; Part II held the frame fixed. Part III turns the generation mechanism into a datatype. Five additions matter.

1. **The pivot is now the diachronic unit.** Definition 2.8 makes precise what Parts I and II only pointed at: the recomputation of the frame, split into extension and rotation, stamped with γ.
2. **Sound change is a theorem, not an image.** Propositions 4.3 and 4.4 give the exact sense in which a sound change is regular (deterministic) and blind (bit-level). The miniature of Table 1 is checkable by hand.

Saussure (Part Three)	In the *Course*	In HLLSet Algebra
diachronic perspective	successive terms, not coexisting	the trajectory read through the transitions $\mathsf{Tr}(t)$: Reading 3.1
diachronic fact	an event	the recorded transition $\mathsf{Tr}(t) = (S(t), H(t-1), \mathsf{Dep}, \mathsf{Ret}, \mathsf{New})$: Definition 2.6
diachronic law	imperative but not general	the recorded transition itself: Proposition 3.3
sound change	absolutely regular, blind	the pivot (extension + rotation): Definitions 2.8; Proposition 4.3
regularity	no exceptions	determinism of the Gaussian step: Proposition 4.4(i)
blindness	no regard for grammar/meaning	the pivot reads bits, never values: Proposition 4.4(ii)
grammatical consequences	bonds broken, alternations	transport $c' = c\mathbf{M}$ breaks/creates coordinate coincidences: Proposition 5.2; alternation = two generation-stamped values: Corollary 5.3
analogy	three terms create a fourth	$\mathsf{Ana}(A, B, C) = A\Delta B\Delta C$, the fourth proportional: Definition 6.2, Theorem 6.3
analogy vs sound change	intelligent vs blind	span event vs basis event: Proposition 6.4
analogy and evolution	conservative and innovative	zero spill, zero residual, new lexicon node: Proposition 7.1; the record decomposes into pivots and in-span commits: Proposition 7.2
folk etymology	incorrect evaluation of an old form	false fourth proportional $D' = \mathsf{Ana}(A, B, C')$ with error $C\Delta C'$: Definition 8.2, Proposition 8.3
agglutination	syntagmatic neighbours merge	promotion of a view to an original (weld $O = A \cup B$): Definition 9.2; pivots iff the meet is unrepresentable: Proposition 9.3
diachronic unit	not the synchronic unit	a path of values across generations: Definition 10.2
diachronic identity	same essence across time	IICA persistence of the bit-set: Proposition 10.3
diachronic reality	the event itself	the recorded transition; objective analysis (provenance + DAG) vs subjective analysis (current frame): Proposition 10.4

Table 2: The complete dictionary of Part Three of the *Course* into HLLSet Algebra.

3. **Analogy is a theorem of GF(2).** Theorem 6.3 is the parallelogram law: the analogical fourth is $A\Delta B\Delta C$, unique, in-span, generation-preserving. Saussure's proportion stops being an observation about speakers and becomes an operation of the ring.
4. **The two forces are type-separated, and a third is located.** Sound change = basis event; analogy = span event (Proposition 6.4); agglutination = view promotion that pivots exactly when the meet is unrepresentable (Proposition 9.3).
5. **The exact pivot trigger.** Proposition 2.9(4) records the exact trigger of a sound change: res $\neq \varnothing$. Positive spill is the visible sufficient sign; the meet obstruction is the invisible case; and the series now states the refined condition uniformly in Parts I and III.

13 Conclusion and outlook to Part IV

Part Three of the *Course* has been re-founded as the theory of the pivot and its two species. Diachronic linguistics is the reading of the trajectory through recorded transitions; sound change is the blind, deterministic pivot; its grammatical consequences are transports of coordinates; analogy is the exact fourth proportional of GF(2); folk etymology is the same operation with a mis-identified term, and the algebra measures the mis-identification; agglutination is the promotion of a view to an original, the syntagmatic source of new oppositions; diachronic units, identities, and realities are paths, IICA persistence, and recorded transitions. Saussure's diachronic linguistics, read in this direction, is no longer a classification of changes: it is the executable theory of a system whose frame moves.

What Part Three deliberately leaves untouched is *where* the changes happen. The plane is one, shared by all speakers; nothing has been said about which communities pivot together, which share a frame, and which do not. Part Four of the *Course*, Geographical Linguistics, is exactly that question: the diversity of languages, the complications of geographical diversity, its causes, and the propagation of linguistic waves. In the algebra these become the theory of worlds: hash configurations and codebooks as dialects, the BSS threshold and the LUT as the boundary of a community, the dispatcher as the speaker who actualizes the shared langue, and the bridge of [8] as the wave that crosses from one world to another. Part IV will re-found geographical linguistics as the multi-world theory of one lattice; Part V will then close the series with retrospective linguistics, in which reconstruction becomes time travel through old frames.

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